

System Requirement Specification (SRS)

System Name: National Youth Bridge Athlete Management System

Objective: To systematically and securely manage youth athlete data, track practice attendance, submit and approve daily leave requests, and manage activity calendars and practice links for athletes and coaches.

1. Gap Analysis & Enhancements

- **Authentication:** Implement Google OAuth 2.0 (Login with Google) for secure identification using athlete and coach emails, which aligns with the system's data structure.
- **Team Management:** Create a dedicated table to manage youth teams (e.g., Under-21, Women's Team) to prevent data duplication and serve as a foreign key for user roles.
- **Automated Practice Link Archiving:** The backend will automatically check the current date. Any Realbridge practice links with past dates will automatically be moved to the "Practice History" section in the UI.

2. Detailed Functional Requirements

2.1 Role-Based Access Control (RBAC)

The system restricts access to menus and functions based on user roles as follows:

Function / Page	Super Admin	Coach	Player
Manage Athlete Data (CRUD)	Manage all teams	View only (Own team)	View/Edit own data only
Attendance Tracking	All teams	Own team only	No access
Leave Request	No access	No access	Submit own requests
Leave Approval	Approve all teams	Approve own team only	No access
Calendar Events	View / Create / Edit	View / Create / Edit	View only
Activity Feed	View / Create / Edit	View only	View only
Practice Links	Manage all links	View Player & Coach links (Own team)	View Player link only (Own team)

2.2 Page Specifications

- **2.2.1 Athlete Profile:** Displays athlete information. Sensitive data is restricted (Phone visible only to owner, Coach, Super Admin; Birth date visible only to owner and Super Admin). Only Super Admin sees the "Create User" button and has full CRUD access. Profile images are uploaded to Google Drive.
- **2.2.2 Attendance Tracking:** For Coaches and Super Admins to record daily practice attendance. Coaches see only their team, while Super Admins can select any team via a dropdown. Options: Present, Absent, Leave.
- **2.2.3 Leave Request:** Allows Players to submit daily leave requests. Submissions default to "Pending". Players can attach evidence (Optional), saved to Google Drive.
- **2.2.4 Leave Approval:** Displays "Pending" leave requests. Coaches can approve/reject for their team; Super Admins can approve/reject any request.
- **2.2.5 Calendar:** Displays activities and practice schedules in a calendar view. All users can view. Coaches and Super Admins can click to create, edit, or delete events.
- **2.2.6 Activity Feed:** Information board for news and activities. All users can view. Only Super Admins can create new feed posts (Title, Date From, Date To, Location, Details).
- **2.2.7 Practice Links:** Central hub for daily Realbridge practice links. Super Admins upload the links. Data masking hides Coach links from Players. Past links auto-archive to "Practice History".

3. Technical Architecture

- **Frontend & Backend Layout:** Angular framework for UI and API handling as a Single Page Application (SPA), with a backend API.
- **Database & Storage Gateway:** Google Sheets API for structured data tables and Google Drive API for file storage (profile images, leave evidence).
- **Containerization:** Managed via Docker and Docker Compose, separating Frontend and Backend containers for stability.

4. Database Schema (Google Sheets)

Table 1: Users (User Information)

Column Name	Data Type	Description
user_id	String (PK)	User ID (e.g., U001)
email	String	User email (for Google Account login)
role	String	User role (Super Admin, Coach, Player)
team_id	String (FK)	Team ID the user belongs to
img_avatar_url	String	Profile image URL on Google Drive
th_prefix	String	Title prefix (Thai)
en_prefix	String	Title prefix (English)

Column Name	Data Type	Description
th_first_name	String	First name (Thai)
en_first_name	String	First name (English)
th_last_name	String	Last name (Thai)
en_last_name	String	Last name (English)
phone	String	Phone number (Restricted access)
birth_date	Date	Date of birth (Restricted access)

Table 2: Teams (Team Information)

Column Name	Data Type	Description
team_id	String (PK)	Team ID (e.g., T001)
team_name	String	Team name (e.g., Under-21, Women's Team)

Table 3: Attendance

Column Name	Data Type	Description
attendance_id	String (PK)	Attendance record ID
date	Date	Practice date
player_id	String (FK)	Player ID
team_id	String (FK)	Team ID
status	String	Attendance status (Present, Absent, Leave)
checked_by	String (FK)	ID of checker (Coach or Super Admin)

Table 4: LeaveRequests

Column Name	Data Type	Description
leave_id	String (PK)	Leave request ID

Column Name	Data Type	Description
player_id	String (FK)	Player ID submitting request
team_id	String (FK)	Team ID
leave_date	Date	Date of leave (day-to-day)
reason	String	Reason for leave
evidence_url	String	Evidence file URL on Google Drive (Optional)
status	String	Approval status (Pending, Approved, Rejected)
action_by	String (FK)	ID of approver (Coach or Super Admin)

Table 5: Events (Calendar)

Column Name	Data Type	Description
event_id	String (PK)	Calendar event ID
title	String	Event title
description	String	Event description
start_datetime	DateTime	Start date & time
end_datetime	DateTime	End date & time
created_by	String (FK)	Creator ID (Coach or Super Admin)

Table 6: Activities (Feed)

Column Name	Data Type	Description
activity_id	String (PK)	Activity feed ID
title	String	Activity title
date_from	Date	Start date
date_to	Date	End date
location	String	Location
details	String	Detailed description

Table 7: PracticeLinks (Realbridge)

Column Name	Data Type	Description
link_id	String (PK)	Practice link ID
practice_date	Date	Practice date
team_id	String (FK)	Assigned Team ID
player_link	String	Realbridge URL for Players
coach_link	String	Realbridge URL for Coaches

5. System Diagrams Guide

- **Use Case Diagram:** Defines system interactions and scopes for each user role to ensure proper access control.
- **Sequence Diagram (for Leave and Attendance):** Illustrates data flow from Angular UI buttons, through API endpoints, to specific Google Sheets cells.
- **Entity Relationship Diagram (ERD):** Shows relational mappings (Primary/Foreign keys) across the 7 Google Sheets tables.